



MIRA

a robotic camera-inspection assistant on the Arduino UNO Q

TEAM JQK · QUALCOMM HACK CHALLENGE 2026 · OFFLINE — PRIVATE — HARDWARE-SAFE

Team JQK

FIVE STUDENTS, ONE ARM.

We wanted a robot arm to do something real — not just wave at the camera.

ROLES.

Firmware & safety interlock · servo control & drivers · XR teleoperation ·
computer vision & ML · product & demo.

ONE RULE WE BUILD BY.

Every feature must run with the Wi-Fi router unplugged.

PHOTOGRAPH — THE TEAM

five of us + the rig
(drop in when captured)

Close-up inspection is manual, hazardous, or cloud-bound

PROBLEM.

Inspecting objects up close means hands, ladders or expensive rigs — and the smart-camera products that exist stream your video to someone else's cloud.

PRODUCT.

Mira: an SO-101 arm carries a camera you steer from a phone in XR. It scans, inspects, records and replays trajectories.

THE TWIST.

Control, capture and the UI itself run on one 2 GB UNO Q. No GPU, no cloud — nothing leaves the device.

FIG. 2 — THE LOOP

PHONE — WebXR UI

Wi-Fi • the board is the access point

UNO Q — QRB2210 • DEBIAN

capture • 100 Hz control • web server

USB serial • 1 Mbaud

SO-101 ARM + WRIST CAM

12 V rail — relay-gated

STM32 SAFETY INTERLOCK

What you will see today

WORKING.

100 Hz servo control · camera capture · trajectory record & replay · fail-safe servo-power interlock.

IN PROGRESS.

XR teleoperation from an Android phone (WebXR + ARCore).

DEMO FLOW.

Arm the interlock → steer from the phone → replay the scan → pull the heartbeat, watch the rail drop.

DEMO STILLs / VIDEO

screen-record + rig footage — pending from team

NOTICE — SAFETY IS A FEATURE

Board power-off does **not** stop the arm — servos freeze rigid. So the MCU alone gates the 12 V rail:

heartbeat stale → **rail drops**

E-stop opens → **rail drops**

corrupt frame → **rail drops**

faults latch → **a human resets**

Linux cannot override it. We demo this live.

From teleop to autonomy — still offline

NOW → HACK DAY.

Finish XR teleop · capture demo footage · harden the USB boot-order workaround.

NEXT MONTH.

On-device 3D reconstruction (visual hull) from arm-swept captures · voice control with on-device speech-to-text.

AFTER.

LeRobot-format datasets from replayed trajectories → train small imitation policies that run on the QRB2210. **The arm learns the inspection, then does it alone.**

Mira — inspection that works anywhere.

Because it never needed anywhere else.

`github.com/Catnip-harvest/Mira-on-Uno-Q` · TEAM JQK · QUESTIONS WELCOME